INCH-POUND
MIL-F-24704/2A(SH)
23 March 1992
SUPERSEDING
MIL-F-24704/2(SH)
30 November 1988

MILITARY SPECIFICATION SHEET

FLANGES, FOUR BOLT SQUARE, BUTT WELD FOR HYDRAULIC SYSTEMS

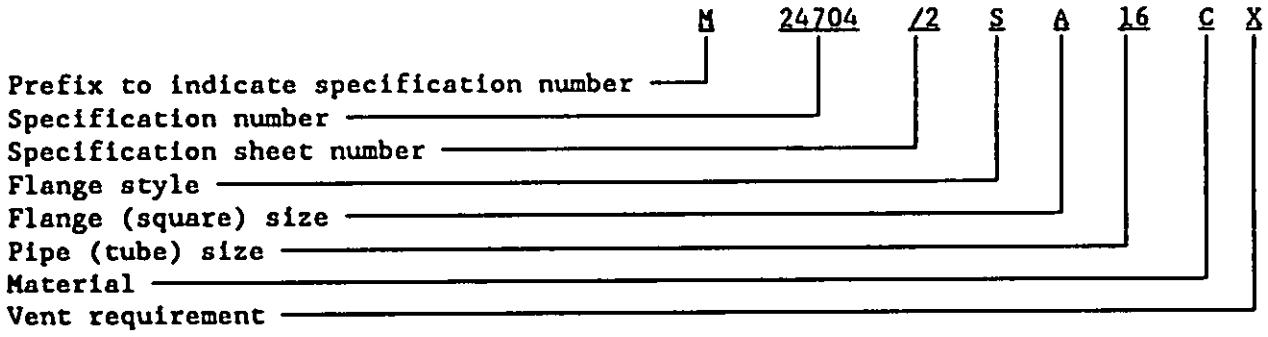
This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-24704.

SCOPE

Scope. This specification sheet covers the requirements for four bolt square butt weld flanges for use in 3000 pounds per square inch (lb/in^2) hydraulic systems.

Classification. Butt weld flanges shall be designated by part identification numbers as identified below:



Example: The part identification number for a butt weld straight flange for nps piping, size 1.5 inches square, 2 inches (nps), made of CRES Class 316, with a vent located at position X (figure 5) would be written M24704/2SA16CX.

AMSC N/A
FSC 4810
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Flange style. The flange style designation letter indicates whether a straight or angle flange and whether for tubing or nominal pipe size (nps) piping as indicated below:

S = Straight flange for nps piping (see figure 1).
L = Right angle flange for nps piping (see figure 2).
T = Straight flange for outside diameter (od) tubing (see figure 3).
R = Right angle flange for od tubing (see figure 4).

Flange (square) size. The square size of the flange shall be designated by a letter in accordance with the following list (see figures 1, 2, 3 and 4 for listing of compatible flange and pipe (tube) sizes):

A = 1.50 inches square.
B = 1.75
C = 2.00
D = 2.25
E = 2.75
F = 3.50
G = 4.25

Pipe (tube) size. The pipe or tubing size shall be designated as a two digit number in accordance with the following list:

Size	Pipe (nps)	Tube (od)
01	1/8 (0.405 od)	---
02	1/4 (.540 od)	0.250
03	3/8 (.675 od)	.375
04	1/2 (.840 od)	.500
06	3/4 (1.050 od)	.750
08	1 (1.315 od)	1.000
10	1-1/4 (1.660 od)	1.250
12	1-1/2 (1.900 od)	1.500
16	2 (2.375 od)	2.000

Material. The flange material shall be designated by a letter as indicated below:

C = Corrosion resistant steel (CRES), class 304 or 316 in accordance with ASTM A 276, ASTM A 473 or QQ-S-763.
L = Corrosion resistant steel (CRES), class 304L or 316L in accordance with ASTM A 276, ASTM A 473 or QQ-S-763.
N = Nickel copper alloy in accordance with QQ-N-281, class A or ASTM B 127 (UNS N04400 only).

Vent requirement. The vent requirement shall be designated by a letter X, Y, or Z as indicated on figures 5 and 6. For flanges without a vent the letter W shall be used. Machining for the vent port shall be as shown on the applicable figure.

Changes from previous issue. Revision letters are not used to denote changes due to the extensiveness of the changes. These changes included extensive changes to the dimensions contained in the tables and to the tolerancing of the associated figures.

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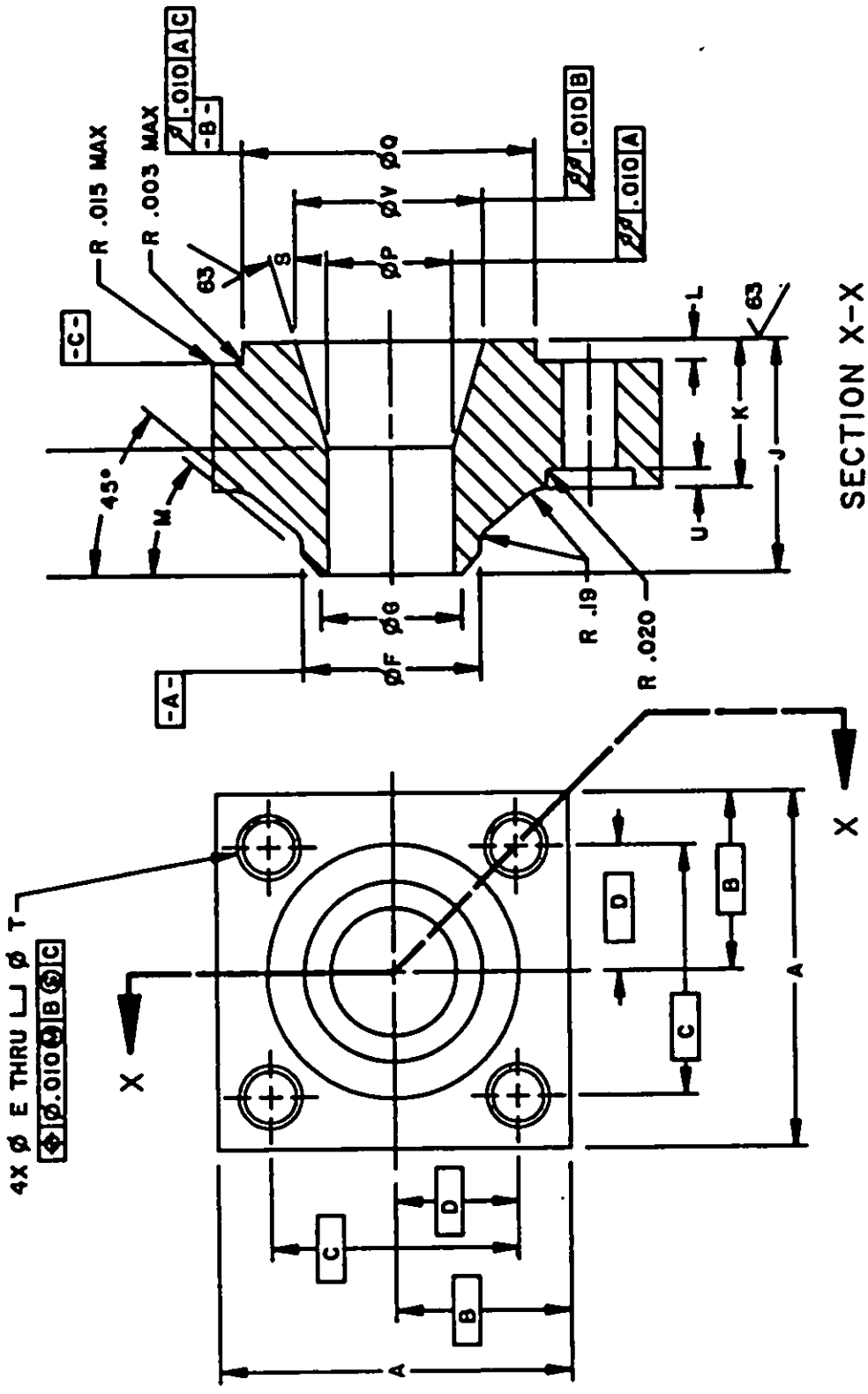


FIGURE 1. Straight flange - pipe.

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UNLESS OTHERWISE SPECIFIED

- Surface roughness shall not exceed 125 microinches.
- Tolerances shall be:
  .XXX +/-.015
  .XX +/-.03
- Break edges and corners to .005 Radius Max.

PART NUMBER --- M 24704/2 S# ** xx
** --- SELECTED FROM BELOW

#	**	FLANGE SIZE (INCH)	PIPE SIZE (NPS)	A SQ	B	C SQ	D	E +.010 -.010	F	G +.010 -.010	J	K
F	08	3.50	1.000	3.50	1.75	2.500	1.250	0.531	1.315	1.160	2.125	1.250
F	10	3.50	1.250	3.50	1.75	2.500	1.250	0.531	1.660	1.301	2.125	1.250
F	12	3.50	1.500	3.50	1.75	2.500	1.250	0.531	1.900	1.520	2.125	1.250
G	16	4.25	2.000	4.25	2.12	3.062	1.531	0.656	2.375	1.698	2.594	1.750

#	**	FLANGE SIZE (INCH)	PIPE SIZE (NPS)	L +.002 -.000	M -0deg +5deg	P +.010 -.000	Q +.000 -.002	V +.000 -.015	S +/-2.5	T	U
F	08	3.50	1.000	0.125	40	0.957	2.124	1.500	15.0	0.915	0.093
F	10	3.50	1.250	0.125	40	1.281	2.124	1.500	5.0	0.915	0.218
F	12	3.50	1.500	0.125	40	1.500	2.124	---	----	0.915	0.218
G	16	4.25	2.000	0.124	40	1.688	2.374	1.750	15.0	1.228	0.500

FIGURE 1. Straight flange - pipe - Continued.

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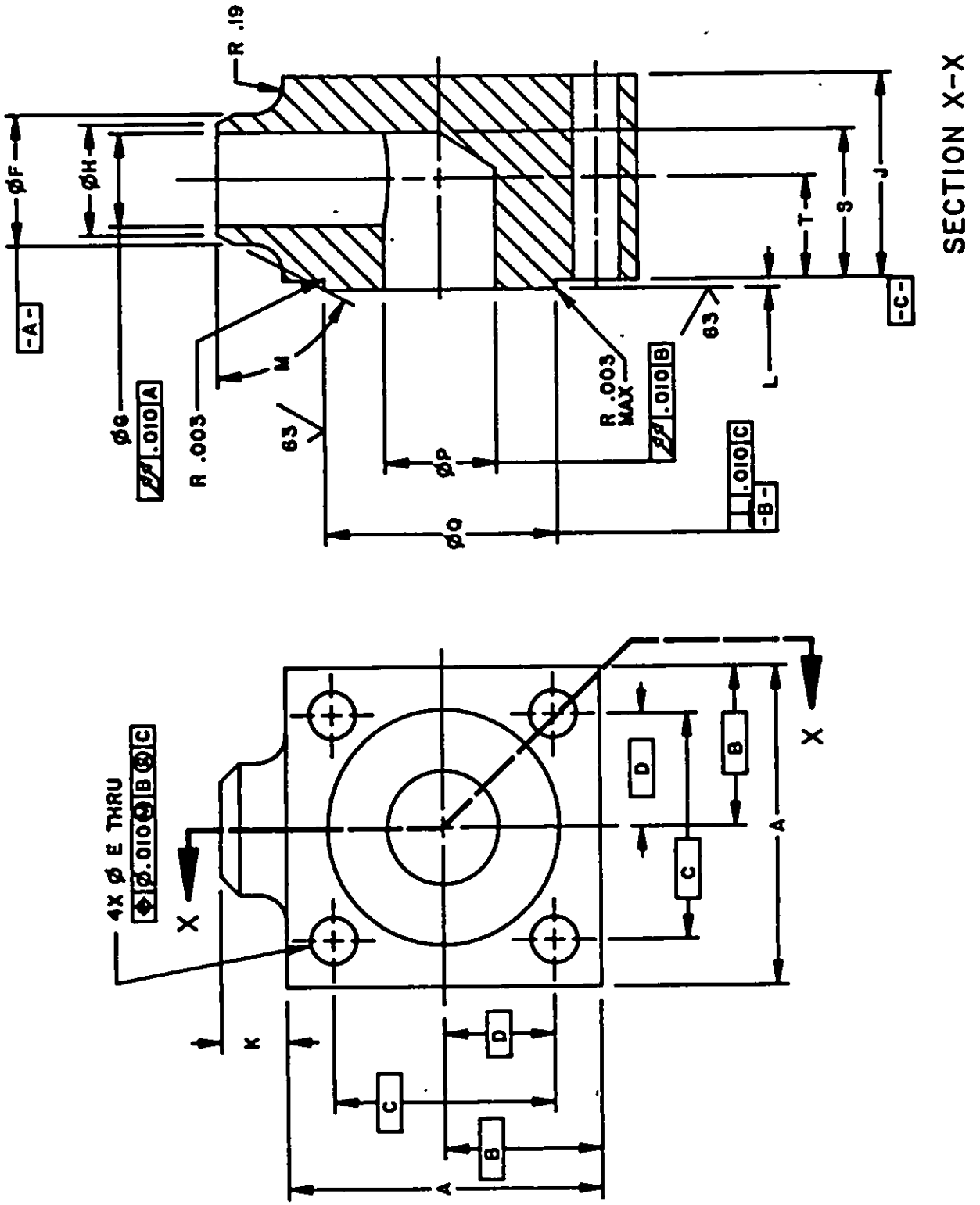


FIGURE 2. Right angle flange - pipe.

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UNLESS OTHERWISE SPECIFIED

- Surface roughness shall not exceed 125 microinches.
- Tolerances shall be:
  .XXX +/-.015
  .XX +/-.03
- Break edges and corners to .005 Radius Max.

PART NUMBER ---- M 24704/2 L# ** xx
** --- SELECTED FROM BELOW

#	**	FLANGE SIZE (INCH)	PIPE SIZE (NPS)	A SQ	B	C SQ	D	E +.010 -.010	F	G +.010 -.010	H +.010 -.010	J	K
F	08	3.50	1.000	3.50	1.75	2.500	1.250	0.531	1.315	0.957	1.160	2.189	0.875
F	10	3.50	1.250	3.50	1.75	2.500	1.250	0.531	1.660	1.281	1.344	2.689	0.875
F	12	3.50	1.500	3.50	1.75	2.500	1.250	0.531	1.900	1.500	1.562	2.939	0.875
G	16	4.25	2.000	4.25	2.12	3.062	1.531	0.656	2.375	1.688	1.750	3.532	1.000

#	**	FLANGE SIZE (INCH)	PIPE SIZE (NPS)	L +.002 -.000	M -0deg +5deg	P +.000 -.015	Q +.000 -.002	T	S MAX
F	08	3.50	1.000	0.125	40	1.500	2.124	0.957	1.656
F	10	3.50	1.250	0.125	40	1.500	2.124	1.312	2.000
F	12	3.50	1.500	0.125	40	1.500	2.124	1.312	2.094
G	16	4.25	2.000	0.125	40	1.750	2.374	1.750	2.469

FIGURE 2. Right angle flange - pipe - Continued.

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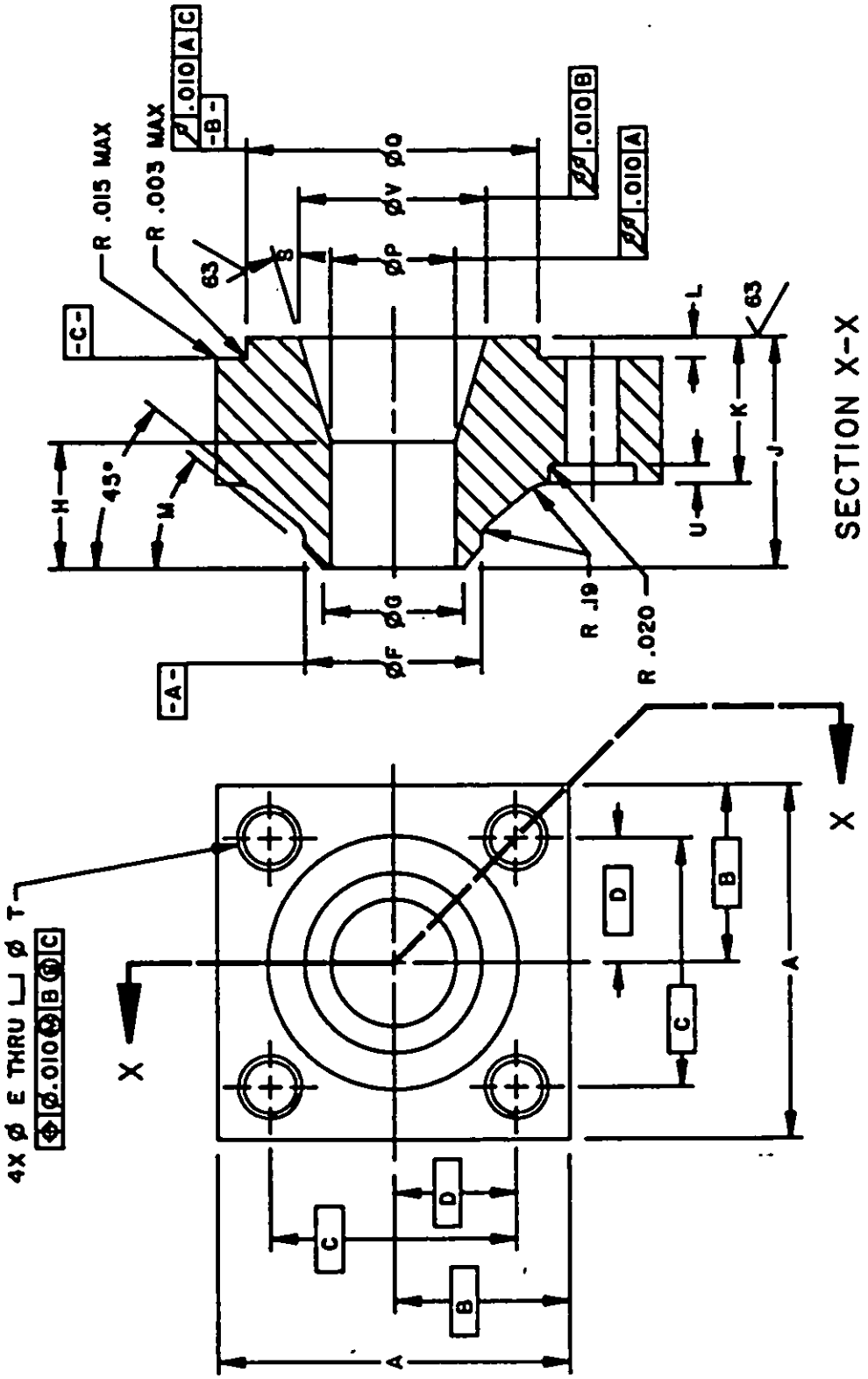


FIGURE 3. Straight flange - tube.

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UNLESS OTHERWISE SPECIFIED

- Surface roughness shall not exceed 125 microinches.
- Tolerances shall be:
  .XXX +/-.015
  .XX +/-.03
- Break edges and corners to .005 Radius Max.

PART NUMBER --- M 24704/2 T# ** xx
** --- SELECTED FROM BELOW

#	**	FLANGE SIZE (INCH)	TUBE SIZE (OD)	A SQ	B	C SQ	D	E +.010 -.010	F	G +.010 -.010	J	K
F	10	3.50	1.250	3.50	1.75	2.500	1.250	0.531	1.250	0.748	2.125	1.250
F	12	3.50	1.500	3.50	1.75	2.500	1.250	0.531	1.500	0.872	2.125	1.250
G	16	4.25	2.000	4.25	2.12	3.062	1.531	0.656	2.000	1.185	2.594	1.750

#	**	FLANGE SIZE (INCH)	TUBE SIZE (OD)	L +.002 -.000	M -0deg +5deg	N	P +.010 -.000	Q +.000 -.002	S +/-2.5 DEG	T	U	V +.000 -.015
F	10	3.50	1.250	0.125	40	0.375	0.688	2.124	30.0	0.915	0.218	1.500
F	12	3.50	1.500	0.125	40	0.375	0.812	2.124	30.0	0.915	0.218	1.500
G	16	4.25	2.000	0.124	40	0.375	1.500	2.374	30.0	1.228	0.500	1.750

FIGURE 3. Straight flange - tube - Continued.

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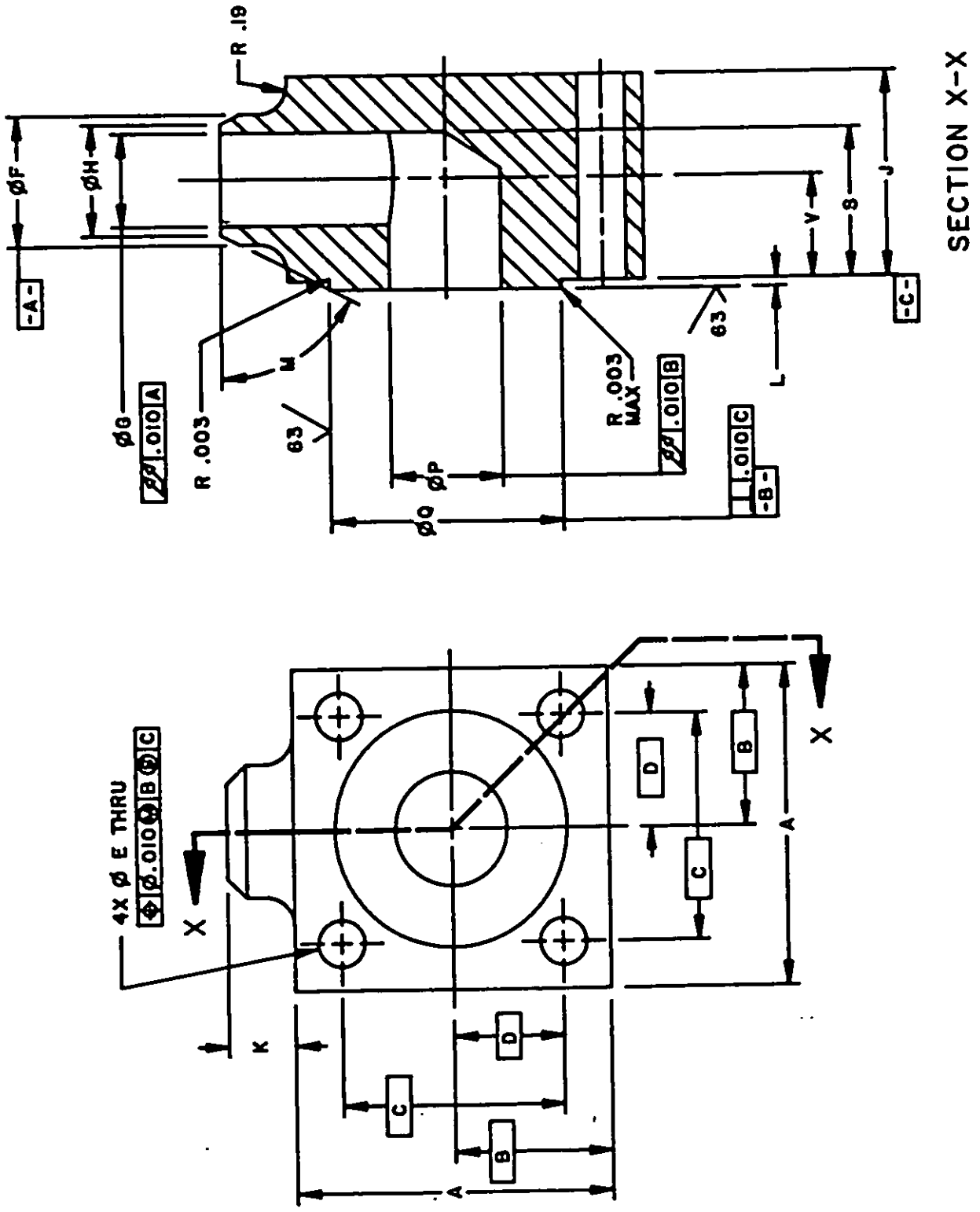


FIGURE 4. Right angle flange - tube.

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- Surface roughness shall not exceed 125 microinches.
- Tolerances shall be:
  .XXX +/-.015
  .XX +/-.03
-Break edges and corners to .005 Radius Max.

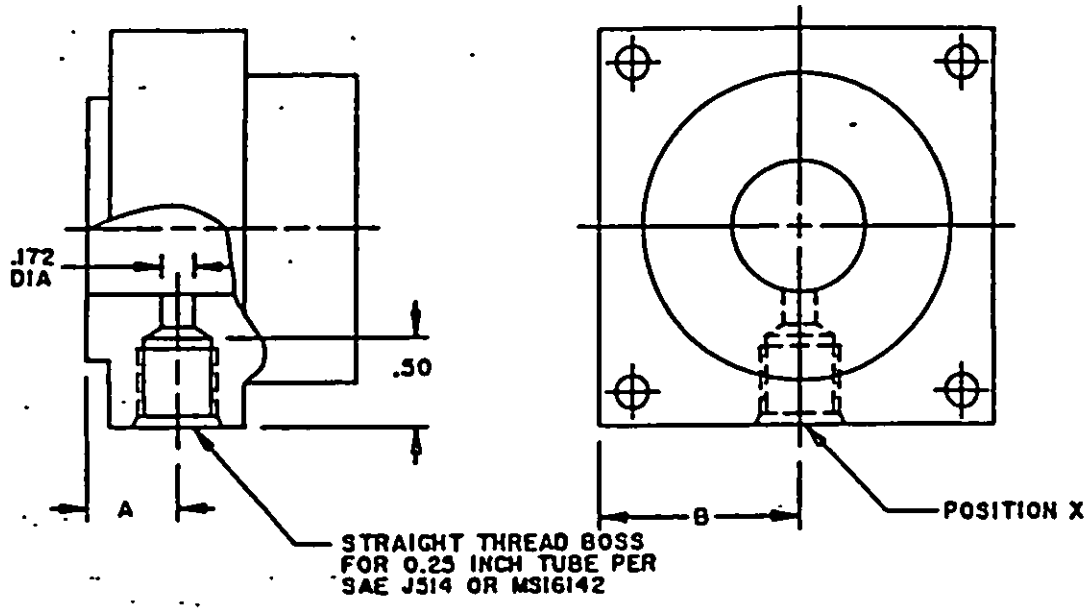
PART NUMBER ---- M 24704/2 R# ** xx
** --- SELECTED FROM BELOW

# **	FLANGE SIZE (INCH)	TUBE SIZE (OD)	A SQ	B	C SQ	D	E +.010 -.010	F	G +.010 -.010	H +.010 -.010	J	K
F 10	3.50	1.250	3.50	1.75	2.500	1.250	0.531	1.250	0.688	0.748	2.689	0.875
F 12	3.50	1.500	3.50	1.75	2.500	1.250	0.531	1.500	0.812	0.872	2.939	0.875
G 16	4.25	2.000	4.25	2.12	3.062	1.531	0.656	2.000	1.125	1.185	3.532	1.000

# **	FLANGE SIZE (INCH)	TUBE SIZE (OD)	L +.002 -.000	M -0deg +5deg	P +.000 -.015	Q +.000 -.002	V	S MAX
F 10	3.50	1.250	0.125	40	0.688	2.124	1.312	1.656
F 12	3.50	1.500	0.125	40	0.812	2.124	1.312	1.718
G 16	4.25	2.000	0.125	40	1.500	2.374	1.750	2.312

FIGURE 4. Right angle flange - tube - Continued.

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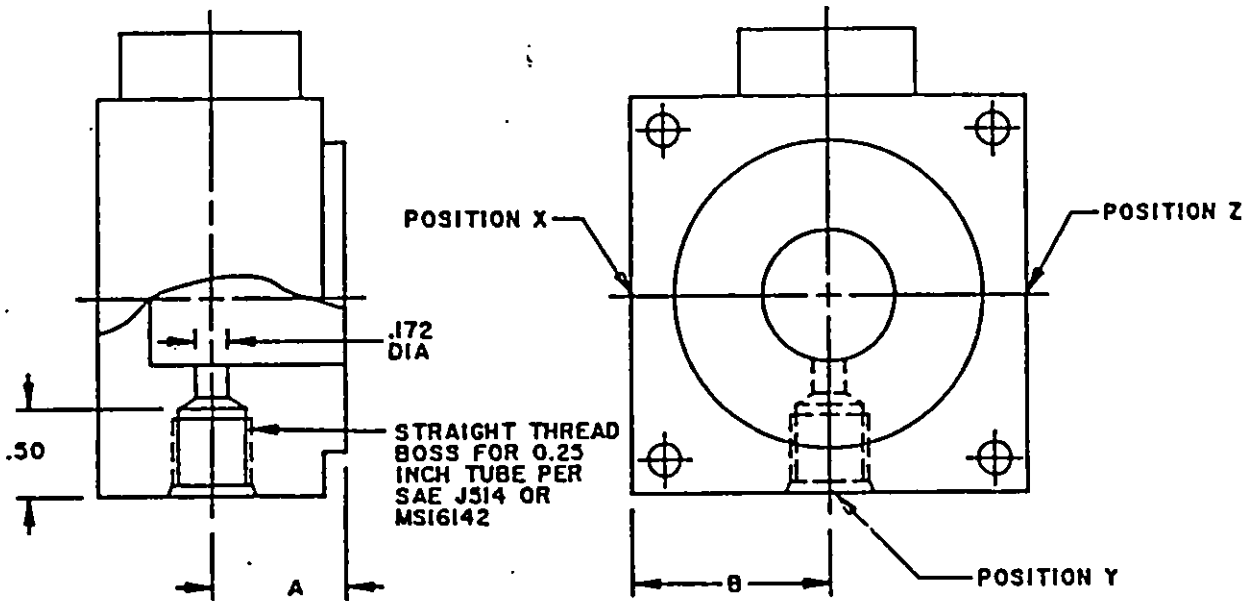
THE VENT PORT CAN BE LOCATED AT POSITION X FOR THE FOLLOWING FLANGES. INSERT THE LETTER X AT THE APPROPRIATE LOCATION IN THE FLANGE PART NUMBER. FOR FLANGES WITHOUT VENT REQUIREMENT, INSERT THE LETTER W.

UNLESS OTHERWISE SPECIFIED
- Tolerances shall be:
  .XXX +/-.015
  .XX +/-.03

#	**	FLANGE SIZE (INCH)	PIPE OR TUBE	A	B
F	08	3.50	1.000	0.672	1.75
F	10	3.50	1.250	0.734	1.75
F	12	3.50	1.500	0.734	1.75
G	16	4.25	2.000	0.906	2.12

FIGURE 5. Straight flange vent location.

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THE VENT PORT CAN BE LOCATED AT POSITION X,Y, OR Z FOR THE FOLLOWING FLANGES. INSERT THE LETTER X,Y, OR Z AT THE APPROPRIATE LOCATION IN THE FLANGE PART NUMBER. FOR FLANGES WITHOUT VENT REQUIREMENT, INSERT THE LETTER W.

UNLESS OTHERWISE SPECIFIED
- Tolerances shall be:
  .XXX +/-.015
  .XX +/-.03

#	**	FLANGE SIZE (INCH)	PIPE OR TUBE	A	B
F	08	3.50	1.000	1.188	1.75
F	10	3.50	1.250	1.438	1.75
F	12	3.50	1.500	1.594	1.75
G	16	4.25	2.000	1.875	2.12

FIGURE 6. Right angle flange vent location.

Preparing activity:
Navy - SH
(Project 4810-N082-2)

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